Literature Review: Text-to-Symbolic Conversion for Legal Statutory Reasoning

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1. Introduction

Traditional legal interpretation and statutory reasoning require specialist knowledge, resulting in excessive legal expenses and a lack of accessibility for many individuals. Automating statutory reasoning with artificial intelligence (AI) and symbolic logic offers a promising approach to reducing legal costs while maintaining accuracy and consistency in legal interpretation.

One of the main challenges in AI-driven legal reasoning is the ability to accurately interpret, apply, and reason about statutory laws. Unlike most language processing tasks, statutory reasoning requires strict logical consistency, exception handling, and significant contextual knowledge. Large Language Models (LLMs), such as GPT-4, perform well in natural language processing but struggle with logical reasoning and the structured application of legal principles. On the other hand, symbolic AI, which is based on formal logic, ensures correct legal interpretation but lacks adaptability and scalability. Hybrid neuro-symbolic techniques aim to close this gap by combining LLMs' textual comprehension with the logical precision of symbolic AI, offering a promising route forward for legal automation.

In this literature review, we explore existing research on statutory reasoning, AI-driven legal interpretation, and hybrid neuro-symbolic models. It examines:

- **Computational Law**: The development of automated systems for legal processes, such as contract analysis and compliance checks.
- **Statutory Reasoning Challenges**: Persistent issues include ambiguities in legal language, conflicts between rules, and the lack of standardized interpretation frameworks.
- **AI Methodologies**: A comparison of symbolic AI (e.g., rule-based expert systems), LLMs (e.g., GPT-4), and hybrid neuro-symbolic architectures that merge logic with machine learning.
- **The SARA Dataset**: A benchmark for evaluating AI systems' statutory reasoning performance, highlighting the importance of logical rigor and real-world applicability.
- **Advancements and Limitations**: While AI-driven models show promise, gaps in consistency, interpretability, and scalability hinder their adoption in high-stakes legal settings.

We will evaluate existing research and methodologies to identify key limitations, gaps, and future directions in the development of AI-driven statutory reasoning models. The goal is to improve the efficiency, accessibility, and affordability of legal services through AI-driven automation, ensuring greater legal consistency and fairness.

2. Background

2.1 Computational Law

Computational law is a new field that focuses on automating legal reasoning and decision-making using formal logic, programming languages, and artificial intelligence (AI). This aims to transform conventional legal procedures into more organized, machine-executable frameworks by enabling automated compliance checking, legal decision support, and digital contract execution (Genesereth 2015). As laws become more complex, computational law provides a more efficient and scalable solution for legal interpretation, reducing human effort while improving consistency in legal applications.

Computational law involves formalizing legal rules into structured representations that allow machines to read and apply them. Unlike traditional legal systems, which heavily rely on human interpretation and discretion, computational law utilizes rule-based logic and automated reasoning to evaluate legal situations (Merigoux, Chataing, and Protzenko 2021). It also integrates formal logic, natural language processing (NLP), and domain-specific programming languages to convert legal statutes into machine-readable formats.

Statutory interpretation is one of the most prominent applications of computational law—it involves translating legal texts into executable logic programs. Catala is a programming language specifically designed for this purpose, allowing laws to be written in a manner that directly corresponds to their legal structure while maintaining computational executability (Merigoux, Chataing, and Protzenko 2021). This enables automatic legal compliance checks and decision-making, reducing reliance on manual legal analysis.

2.2 Statutory Reasoning

2.2.1 Key Concepts of Statutory Reasoning

Statutory reasoning is the process of interpreting and applying laws written by legislative bodies to specific cases. Unlike case law, which relies on judicial precedents, statutory reasoning requires strict adherence to legal texts, such as statutes, regulations, and codes. The primary challenge lies in the precise interpretation of legal provisions, which often contain complex structures, exceptions, and cross-references (Holzenberger 2022; Blair-Stanek, Holzenberger, and Durme 2023; Blair-Stanek, Holzenberger, and Van Durme 2023).

A key aspect of statutory reasoning is ensuring that legal rules are applied consistently. Lawyers and judges engage in textual analysis, contextual interpretation, and logical deduction to determine how statutes apply to real-world situations. However, ambiguities in statutory language, evolving legal frameworks, and conflicting provisions make this process inherently difficult (Genesereth 2015).

2.2.2 Dataset for Statutory Reasoning - SARA

Blair-Stanek, Holzenberger, and Durme (2023)'s research has shown that even advanced AI models struggle with basic statutory interpretation, highlighting the complexity of this task. The Statutory Reasoning Assessment dataset (SARA) is a benchmark designed to evaluate AI models' ability to perform formal statutory reasoning—interpreting and applying legal rules to specific factual scenarios, a key challenge in AI-driven legal analysis (Holzenberger, Blair-Stanek, and Durme 2020).

SARA consists of legal statutes written in controlled natural language, derived from the U.S. Tax Code, along with corresponding factual cases that require formal reasoning. It provides a structured environment to assess whether models can understand, interpret, and reason with statutory language. Given a statute and a factual scenario, the AI must determine whether a particular outcome follows from the legal text. AI models are evaluated across varying difficulty levels, ranging from simple rule application to complex multi-step reasoning (Holzenberger, Blair-Stanek, and Durme 2020).

SARA v2 extends the original dataset by further breaking down statutory reasoning into fundamental language understanding challenges, enabling a more detailed assessment of AI capabilities(Holzenberger and Durme 2021). It introduces tasks that evaluate logical consistency, ambiguity resolution, rule combination, and negative reasoning. Logical consistency tests whether AI models correctly follow the structure of legal rules, while ambiguity resolution examines their ability to interpret context-dependent statutory phrases. Rule combination assesses performance when multiple legal provisions interact, and negative reasoning introduces cases where the absence of a specific clause affects the outcome (Holzenberger and Durme 2021).

3. AI Reasoning Methods

3.1 Symbolic AI Approaches

Symbolic AI is a logic-based approach to reasoning that represents laws in a structured and unambiguous manner using formal logic, such as first-order logic (FOL) (Nguyen et al. 2022). Unlike natural language, which is inherently ambiguous and constantly evolving, symbolic representations ensure clarity, consistency, and interpretability in logical applications (Borazjanizadeh and Piantadosi 2024; Sun et al. 2024).

Benzmüller, Parent, and Torre (2020) developed a framework based on higher-order logic (HOL) that uses symbolic logic to represent and analyze legal rules. It leverages off-the-shelf theorem provers and model finders to enhance reasoning performance. Unlike statistical machine learning models, this symbolic reasoning framework ensures explainability and logical consistency.

3.1.1 Symbolic AI in Legal Statutory Reasoning

Symbolic AI is particularly useful in statutory reasoning, where legal texts must be translated into a structured format that allows for logical inference. Borazjanizadeh and Piantadosi (2024) found that AI systems can apply deductive reasoning to determine whether a law applies to a given case when both statutes and case facts are represented in formal logic. By structuring legal rules and case details in a logical framework, these systems strictly follow predefined rules to reach conclusions. This approach ensures consistency and interpretability, reducing ambiguity in statutory applications.

Holzenberger and Van Durme (2023) and Nguyen et al. (2022) propose translating legal texts into structured logical representations, such as Prolog knowledge bases, by combining manual rule formalization with automated methods to improve the accuracy of legal reasoning systems. Since symbolic AI is rule-driven and deterministic, it is particularly suitable for statutory applications where logic is strictly defined.

Another application of symbolic statutory reasoning is the system based on the Japanese JUF theory ("theory of presupposed ultimate facts"), which aids judicial decision-making under incomplete information. This system adapts logic programming-based knowledge representation to improve interpretability for legal professionals (Satoh et al. 2010).

3.1.2 Limitations of Symbolic AI

Despite its strengths, symbolic AI has several limitations. Legal language often includes vague or undefined terms, which symbolic AI struggles to interpret without explicit rules (Genesereth 2021). Additionally, symbolic systems require significant manual effort to build and maintain knowledge bases, making scalability a challenge, as seen in the creation of the SARA dataset (Holzenberger, Blair-Stanek, and Durme 2020). Adapting to new legislation is not automatic, and constructing structured legal knowledge representations, such as knowledge graphs, remains complex and expertise-dependent (Zhong et al. 2020).

While Prolog-based symbolic AI excels in deductive reasoning, it lacks mechanisms for implicit reasoning, limiting its ability to infer unstated but contextually relevant information (Borazjanizadeh and Piantadosi 2024). Additionally, some formal logic systems, such as higher-order logic (HOL), face theoretical challenges like undecidability, which impact their practical usability (Benzmüller, Parent, and Torre 2020).

3.2 Reasoning in LLMs

Large Language Models (LLMs) are deep learning models designed to process and generate human-like text based on statistical patterns. Unlike symbolic AI, which uses explicit logic and structured rule systems, LLMs rely on probabilistic learning to determine relationships between words and concepts (Katz et al. 2023; Janatian et al. 2023). They are mainly good at contextual understanding, text generation, and pattern recognition, which makes them extremely useful for tasks like legal document analysis, summarization, and preliminary statutory interpretation (Chalkidis et al. 2020; Bench-Capon and Atkinson 2022).

Though they excel in these areas, LLMs also suffer from structured reasoning problems and lack logical consistency, especially when it comes to applying statutory law. They often face errors when dealing with complex legal logic due to their heavy reliance on pattern matching rather than explicit rule-following (Blair-Stanek, Holzenberger, and Durme 2023; Holzenberger 2022; Blair-Stanek, Holzenberger, and Van Durme 2023; Katz et al. 2023). Interest in hybrid AI techniques that blend formal rule-based reasoning with LLMs has increased as a result of these limitations.

3.2.1 Advancements in LLM Reasoning: The Rise of Large Reasoning Models (LRMs)

Recent developments in Large Reasoning Models (LRMs) have greatly improved AI's ability to perform structured reasoning tasks. Unlike traditional LLMs, LRMs use reinforcement learning techniques to generate longer and more structured chains of reasoning, making them better suited for tasks requiring logical inference and multi-step deduction (OpenAI 2024a).

One of the most significant advancements in this area is OpenAI's *o1* model, which leads the transition from LLMs to LRMs. To enhance logical consistency, the model employs reinforcement learning from human feedback (RLHF), enabling AI to produce more structured reasoning paths and solve complex tasks effectively (OpenAI 2024a; OpenAI 2024b).

Following the release of *o1*, OpenAI later introduced its *03-mini* model, which achieved state-of-the-art performance on reasoning benchmarks such as Frontier Maths and the ARC-AGI challenge (OpenAI 2025; OpenAI 2024b; Jones 2025). Although OpenAI has not publicly released technical details, research indicates that these models significantly outperform traditional LLMs in organized legal reasoning tests.

Beyond OpenAI, DeepSeek-R1 has successfully replicated the performance of *o1*, providing a more transparent research framework for understanding reinforcement learning-based reasoning models (DeepSeek-AI et al. 2025). Similarly, *Kimi k1.5* has demonstrated reinforcement learning techniques for structured reasoning, reinforcing the shift toward LRMs for legal AI applications (Kimi 2025).

3.2.2 Challenges and Limitations of LLMs in Statutory Reasoning

While LRMs mark a significant improvement over traditional LLMs, applying them to legal reasoning presents several challenges:

- **Lack of Deep Legal Understanding**:
  LLMs can analyze legal text but often fail to grasp the deeper logic of statutory law. They struggle with intricate legal reasoning tasks, particularly when required to apply specific legal rules to new fact patterns (Yuan et al. 2024).

- **Hallucinations and Inaccuracies**:
  A major issue with LLMs is their tendency to generate believable but inaccurate statements. In legal contexts, where precision is crucial, this can lead to severe misinterpretations. Research has shown that LLMs frequently produce factually incorrect legal claims, and their own confidence levels fail to predict when they are making errors (Dahl et al. 2024).

- **Inability to Apply Unseen Statutes**:
  LLMs rely on training data, meaning they often fail to generalize legal rules they have not encountered before. This poses a significant problem in dynamic legal environments, where laws frequently change (Blair-Stanek, Holzenberger, and Durme 2023; Holzenberger 2022; Blair-Stanek, Holzenberger, and Van Durme 2023).

3.3 Neuro-Symbolic Reasoning Approaches

Moving forward, researchers are exploring ways to integrate these models with formal logic frameworks to improve legal reasoning consistency and interpretability. One of them is neuro-symbolic reasoning, which integrates symbolic AI (explicit rules, logic) with neural networks (deep learning, pattern recognition).

3.3.1 Neuro-Symbolic: A Hybrid Method

To address the limitations of traditional symbolic AI, particularly its lack of scalability and inability to handle implicit reasoning, researchers have developed hybrid approaches that integrate symbolic AI with neural networks.

This approach leverages neural networks to process unstructured input (e.g., natural language) while employing symbolic reasoning to enforce logical consistency (Sun et al. 2024; Borazjanizadeh and Piantadosi 2024). By bridging the gap between flexible learning and structured legal reasoning, neuro-symbolic AI presents a promising approach for enhancing AI reasoning (Chaudhury et al. 2023; Yu et al. 2023).

Neuro-symbolic AI also enhances Large Language Models (LLMs) by improving generalizable and reliable reasoning, especially for explicit deductive and non-linear tasks (Kant et al. 2024). By combining pattern recognition with logical consistency, it enables more interpretable and structured AI reasoning (Mirzadeh et al. 2024; Bougzime et al. 2025).

One approach is Inductive Logic Programming (ILP), which learns interpretable symbolic rules from data, allowing for human inspection and debugging (Chaudhury et al. 2023). Unlike black-box deep learning, ILP provides explicit reasoning structures, improving reliability in mathematical and logical tasks (Borazjanizadeh and Piantadosi 2024).

Recent advances include NESTA (NEuro-Symbolic Textual Agent), which integrates neural semantic parsing with symbolic planning, achieving superior generalization in text-based reinforcement learning (Chaudhury et al. 2023). Additionally, Prolog-enhanced LLMs improve mathematical reasoning by tackling complex tasks in the Non-Linear Reasoning (NLR) dataset (Borazjanizadeh and Piantadosi 2024).

A major milestone is AlphaGeometry, an AI that solves Olympiad-level geometry problems by combining a neural model with a symbolic deduction engine. It successfully solved 25 out of 30 competition problems, demonstrating the effectiveness of neuro-symbolic reasoning in structured problem-solving (Trinh et al. 2024).

3.3.2 Neuro-Symbolic for Legal AI Applications

In the study by Bhuyan et al. (2024), neuro-symbolic AI has been successfully applied to various domains, including question answering, robotics, computer vision, and healthcare. However, its applications and research in legal AI remain relatively limited.

Prolog has been used to encode laws into first-order logic programs, enabling automated statutory reasoning. Hybrid approaches integrate LLMs with logic programming to automatically generate logical representations of legal statutes and structured case details, improving both interpretability and precision in legal decision-making (Kant et al. 2024).

Moreover, Legal Information Extraction (IE) plays a crucial role by mapping unstructured legal text into structured representations for automated reasoning (Holzenberger and Van Durme 2023). Sun et al. (2024) developed NS-LCR (Neuro-Symbolic Legal Case Retrieval), which improves the explainability of legal case retrieval by integrating case-level and law-level logic rules, thereby enhancing ranking performance and supporting multiple retrieval models.

Pulicharla (2025) explored a novel hybrid framework that integrates neural networks with symbolic reasoning, demonstrating its potential for legal research. This approach combines neural document analysis with symbolic reasoning to derive logical arguments, showing promise for applications in healthcare, finance, and education.

4. Challenges and Limitations in Current Studies

Despite advancements in neuro-symbolic AI and logic-based reasoning, existing studies still face significant limitations in applications in statutory reasoning and legal AI systems.

4.1 Dependence on Structured Representations & Rule Formalization

Neuro-symbolic models rely on structured representations such as Abstract Meaning Representation (AMR) and Prolog knowledge bases, which are constrained by predefined vocabularies (e.g., PropBank semantic roles) (Kant et al. 2024). For specialized domains like law and finance, additional domain-specific retraining is required (Ahn et al. 2024). Prolog-based legal reasoning still depends on manual rule encoding by experts, as fully automating legal rule extraction remains unsolved (Borazjanizadeh and Piantadosi 2024; Blair-Stanek, Holzenberger, and Van Durme 2023). Semantic parsing limitations affect the accuracy of converting legal text into structured representations, where small errors can lead to catastrophic failures in logical inference (Holzenberger and Van Durme 2023; Holzenberger, Blair-Stanek, and Durme 2020).

4.2 Computational & Generalization Challenges in Domain Knowledge

Neuro-symbolic models struggle with out-of-domain generalization, especially in legal case retrieval, text-based games, and mathematical reasoning, where new rules and exceptions frequently emerge (Chaudhury et al. 2023; Holzenberger, Blair-Stanek, and Durme 2020).Legal and mathematical reasoning tasks remain brittle, as models often produce inconsistent outputs for reworded or adversarially modified questions (Borazjanizadeh and Piantadosi 2024). Even advanced LLMs (e.g., GPT-4) fail on constraint-based reasoning tasks despite excelling on standard benchmarks (Kant et al. 2024).

4.3 Reliance on Token Bias Instead of True Logical Inference

Empirical studies suggest that LLMs may mimic reasoning rather than actually applying logical inference, relying on statistical token biases instead (Kant et al. 2024).Techniques like chain-of-thought prompting and in-context learning often lead to semantic shortcuts, where models generate plausible-sounding but logically inconsistent outputs (Chaudhury et al. 2023; Mirzadeh et al. 2024).This raises concerns about whether neuro-symbolic approaches truly improve reasoning or if they merely make LLMs appear more interpretable without fundamentally enhancing their logical capabilities.

In summary, while neuro-symbolic AI improves upon symbolic AI in adaptability and reasoning, challenges remain in automating legal rule extraction and preventing models from relying on token bias rather than true logical inference. Addressing these limitations requires further research in representation learning and deeper integration of structured reasoning within LLMs.

5. Conclusion

This literature review has explored the current state of AI-driven statutory reasoning, highlighting the strengths and limitations of symbolic AI, LLMs, and neuro-symbolic approaches. While symbolic AI ensures logical consistency, it struggles with scalability. LLMs, on the other hand, are powerful in natural language understanding but lack structured reasoning capabilities. Neuro-symbolic AI presents a promising hybrid approach, but further research is needed to improve logical consistency, adaptability, and interpretability in legal applications.

Existing research shows that current AI models still struggle to apply legal laws correctly and handle complex legislative interpretations. Future research should focus on creating more robust hybrid models, improving evaluation benchmarks such as SARA, and ensuring that AI-generated legal reasoning is both explainable and dependable.

By integrating insights from computational law, machine learning, and symbolic reasoning, this research aims to contribute to enhancing AI's ability to process, interpret, and apply statutory laws effectively, ultimately improving efficiency and accessibility in legal services.

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7. Gen AI Tools Used

- **Grammarly AI** (`https://www.grammarly.com/`): I used this tool while writing my literature review to check spelling and ensure grammatical accuracy. After completing our parts of the literature review, I reviewed the entire document to find any spelling or grammar mistakes, which Grammarly automatically detected.
- **ChatGPT-4o** (`https://chatgpt.com/`): We used ChatGPT to review our literature review and correct any errors, such as spelling, grammar, or missing citations. Additionally, we utilized it at the start of our project to obtain a guideline for headings and essential information. The following prompts were used:
  - “*Provide me with a literature review guideline for an academic paper. We are doing our practicum project on Text-to-Symbolic Conversion for Legal Statutory Reasoning.*”
  - “*Please review our literature review, mark it from 0-100% based on how well it is written and the information it provides. This is an academic paper, so please be strict with it. Check for any spelling mistakes, grammar mistakes, or missing elements.*”

  - "*How do I improve my literature review based on the reviews you have given me? Please be detailed and explain why and what to improve on. Do I need more papers to read, and by how much could I improve my score?*"
  - "*Please check the grammar and expression, to make it natural, and don't make big changes.*"
  - "*Remove duplications in my content.*"
  - "*Check my LaTeX code.*"

- **DeepSeek** (`https://www.deepseek.com/`): I used DeepSeek to compare results with ChatGPT and gain insights from both models. The following prompts were used:

  - "*Give me guidelines on how I should start a literature review, please give your reasoning.*"
  - "*Please review my Literature Review. Make sure you are strict and mark it from 0-100, following strict guidelines. Check for spelling mistakes, grammar mistakes, anything that could be improved, and how I should go about it.*"
  - "*Please re-review the changes made and re-rank it.*"